

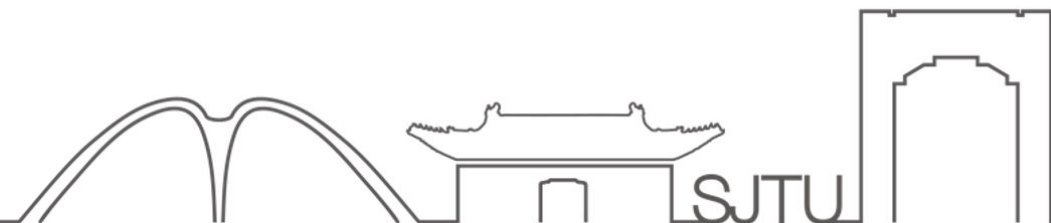


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Angular Momentum

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What to Do When You Get the Exam Problems?



- Read the problem description carefully and identify the information that might be useful.
- Interpret the information that you identified using physical and mathematical language. (Most points will be given for this part!)
 - You can think about:
 - theorems that you learnt in the lecture.
 - model that you learnt through lecture, homework, RC
 - information provided by the problem.
 - tricks provided in RC.
- Synthesize your interpretation and get the final result. (If you cannot solve the whole question, sometimes skip the question might be a wise choice.)

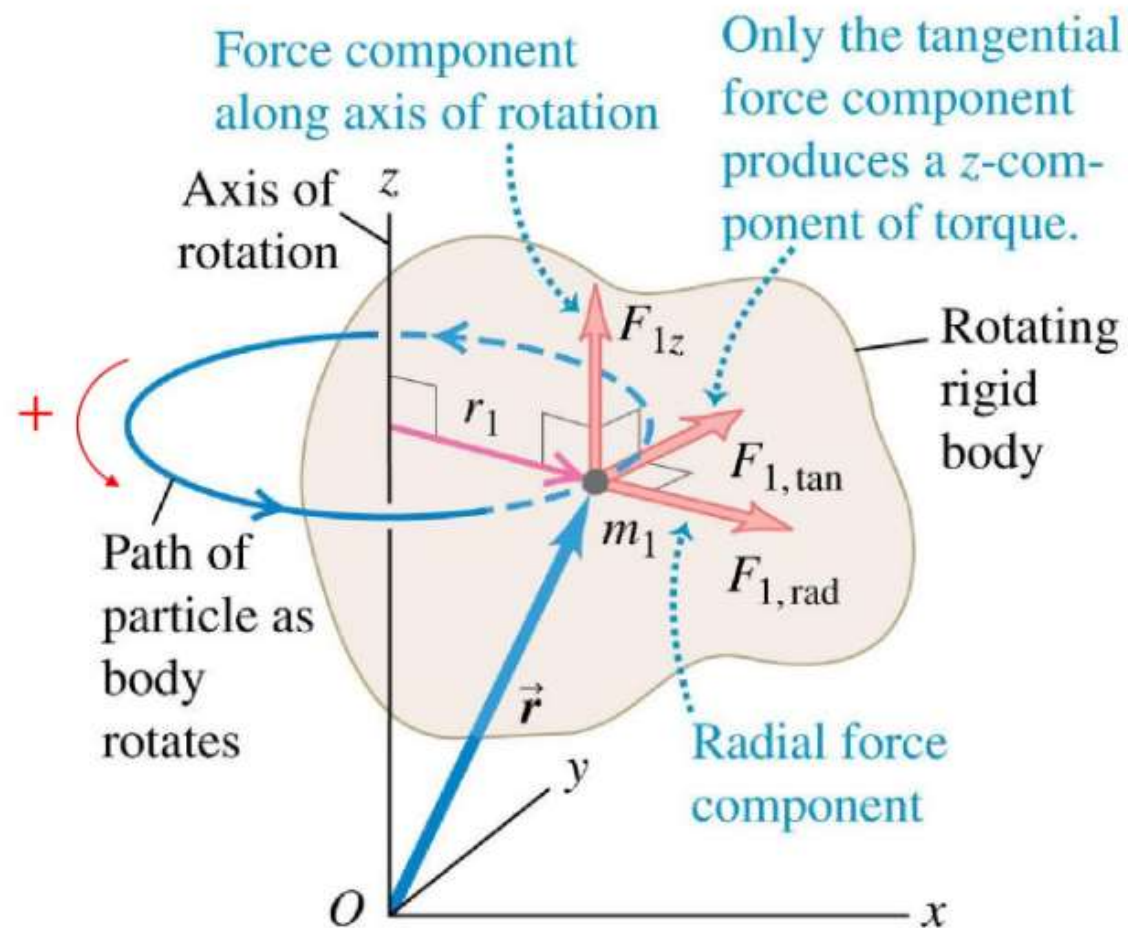
How to get full mark?



- State your solution clearly!!! (Which letter stands for which quantity? Which FoR you use? Which coordinate system you use?)
 - Consider all the possible conditions. (Moving upwards/downwards?)
 - Do not take everything for granted. (Newton's third law.)
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- The exam of VP 150 is not as difficult as you think. You have be calm.

Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$



Angular momentum



Properties:

$$\vec{L} = \vec{r} \times \vec{p} \quad \vec{\tau} = \frac{d\vec{L}}{dt}$$
$$|\vec{L}| = \left| \sum_{i=1}^n m_i \vec{r}_i \times \vec{v}_i \right| = \left(\sum_{i=1}^n m_i r_{\perp i}^2 \right) \omega = I \omega$$

Compared with kinetic energy

$$K = \frac{1}{2} \sum_{i=1}^n m_i |\vec{v}_i|^2 = \frac{1}{2} \left(\sum_{i=1}^n m_i r_{\perp i}^2 \right) \omega^2 = \frac{1}{2} I \omega^2$$

Angular momentum



Considering translational motion:

$$\vec{L} = \sum_{i=1}^n m_i (\vec{r}_c + \vec{r}_i) \times (\vec{v}_c + \vec{\omega} \times \vec{r}_i) = M\vec{r}_c \times \vec{v}_c + I_c \vec{\omega} = \vec{L}_c + \vec{L}'$$

Compared with kinetic energy

$$\begin{aligned} K &= \frac{1}{2} \sum_{i=1}^n m_i |\vec{v}_c + \vec{v}_i - \vec{v}_c|^2 = \frac{1}{2} \sum_{i=1}^n m_i (|\vec{v}_c|^2 + |\vec{v}_i - \vec{v}_c|^2 + \vec{v}_c \cdot (\vec{v}_i - \vec{v}_c)) \\ &= \frac{1}{2} m |\vec{v}_c|^2 + \frac{1}{2} \sum_{i=1}^n m_i |\vec{v}_i - \vec{v}_c|^2 = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2 \end{aligned}$$

Angular momentum



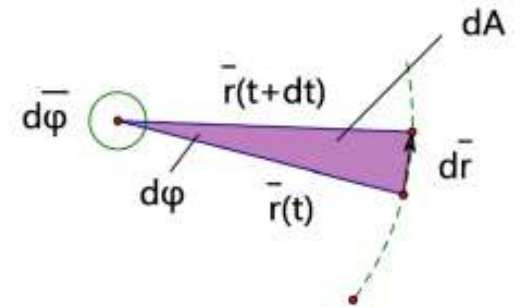
Conservation of Angular momentum: if $\vec{\tau} = 0$, $\frac{d\vec{L}}{dt} = 0$

Example: A planet is moving around the sun. (Central force)

The surface area swept by $\vec{r}$ over dt is $dA = \frac{1}{2} |\vec{r} \times d\vec{r}|$

$$\frac{dA}{dt} = \frac{1}{2} \left| \vec{r} \times \frac{d\vec{r}}{dt} \right| = \frac{dA}{dt} = \frac{1}{2} |\vec{r} \times \vec{v}|$$

As, $\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} = \text{const}$, $\frac{dA}{dt} = \text{const}$



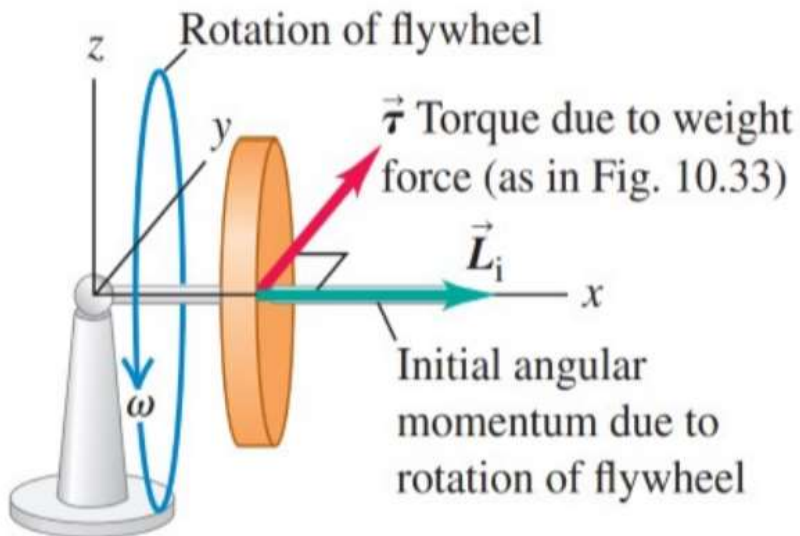
The Gyroscopic effect/Precession (Not in Exam!)



Working Principle

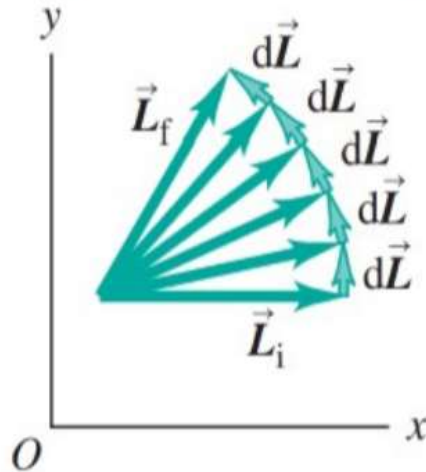
(a) Rotating flywheel

When the flywheel is rotating, the system starts with an angular momentum $\vec{L}_i$ parallel to the flywheel's axis of rotation.



(b) View from above

Now the effect of the torque is to cause the angular momentum to precess around the pivot. The gyroscope circles around its pivot without falling.



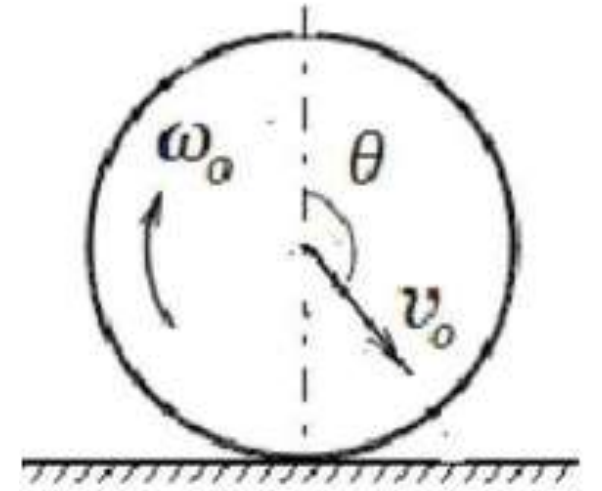
$$\left| \frac{d\vec{L}}{dt} \right| = \frac{d\phi}{dt} I \omega = mgr$$

The angular velocity of the axis

$$\frac{d\phi}{dt} = \frac{mgr}{I\omega}$$

EX:

A solid ball hit the ground (with ω_0 , θ , v_0 shown in the picture), the collision is elastic, the friction coefficient is large enough (no slip). Radius is r
Calculate the final velocity and angular velocity after collision.



Hint: vertical speed has the same magnitude and opposite direction after collision.



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Consider the horizontal speed. $v_0 \sin \theta$ to v
Angular velocity. ω_0 to ω

Conservation of angular momentum

$$mv_0 \sin \theta r + \frac{2mr^2}{5} \omega_0 = mvr + \frac{2mr^2}{5} \omega$$

Conservation of energy

$$\frac{1}{2} m (v_0 \sin \theta)^2 + \frac{mr^2}{5} \omega_0^2 = \frac{1}{2} mv^2 + \frac{mr^2}{5} \omega^2$$



$$\omega = \frac{10v_0 \sin\theta}{7r} - \frac{3}{7}\omega_0 \text{ (backwards)}$$

$$v = v_0 \sin\theta + \frac{4}{7}\omega_0 r - \frac{4v_0 \sin\theta}{7} \text{ (horizontal)}$$

$$v_{\perp} = -v_0 \cos\theta \text{ (vertical)}$$



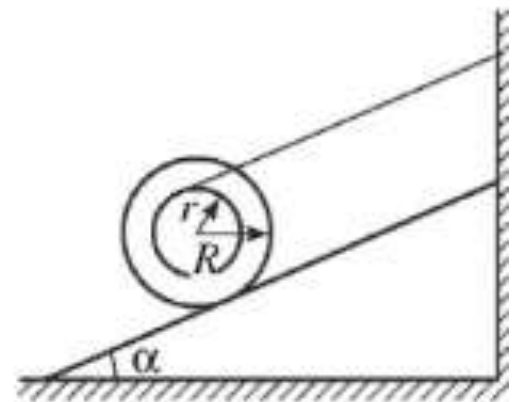
EX

A light inextensible string is wound onto a reel with mass m , inner radius r , and outer radius R placed on a ramp inclined at an angle α to the horizontal. The other end of the string is attached to a wall, so that the string is parallel to the plane. The string does not slip on the reel.

What is the acceleration of the center of mass of the reel?

The moment of inertia of the reel about the axis of symmetry is I_0 and the coefficient of kinetic friction between the reel and the incline is μ , and the acceleration due to gravity is g .

Hint. To analyse the rotational motion of the reel, you may choose the axis through the string's point of contact with the reel.



$$\frac{d\omega}{dt} (I_0 + mr^2) = mgr \sin \alpha - \mu mg \cos \alpha (R + r)$$

$$\frac{d\omega}{dt} = \frac{mgr \sin \alpha - \mu mg \cos \alpha (R + r)}{I_0 + mr^2}$$

if $\mu < \tan \alpha \frac{r}{R+r}$

$$a = r \frac{d\omega}{dt} = \frac{mgr^2 \sin \alpha - \mu mg \cos \alpha (R + r)r}{I_0 + mr^2}$$

else $a = 0$

direction: along the slope, downwards.



References

1. Young, Hugh D., and Roger A. Freedman. University Physics: Volume 1. 15th ed., Global Edition, Pearson, 2020.
2. VP150 SU23 Final RC, Huang Jiahe
3. VP150 SU23 Final RC, Liu Yening
4. VP150 SU23 RC 5, Zhou Sikuang
5. VP150 SU24 Lecture Slide, Madjid Mesli

Thank You!

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